

①

	Annual revenue	Net Profit
Acme Computer	138	8
Nadir Software	87	5
Quixote Media	111	13

a) Quixote had largest annual revenue

Ⓕ

b) Nadir had lowest net profit and Acme Computer had largest annual revenue

Ⓓ

c) Acme Computer had the largest net profit or Quixote had the largest net profit

Ⓓ

d) If Quixote had the smallest net profit then Acme had the largest annual revenue

Ⓓ

-?

e) Nadir had the smallest net Profit if and only if Acme had the largest annual rev



② P — It is below freezing

q — It is snowing

a) It is below freezing and snowing.

$P \wedge q$ (P and q)

b) It is below freezing but not snowing

$P \wedge \neg q$

c) It is not below freezing and it is not snowing

$\neg P \wedge \neg q$

d) It is either snowing or below freezing

$q \vee P$



e) If it is below freezing it is also snowing
 $P \rightarrow q$

f) Either it is below freezing or it is snowing
but it is not snowing if it is below freezing
 $P \oplus q$ -? $(P \vee q) \wedge (P \rightarrow \neg q)$

g) That it is below freezing is necessary
and sufficient for it to be snowing
 $q \leftrightarrow P$ -? $P \leftrightarrow q$

③ a) $P \rightarrow \neg P$ 2^n $2^1 = 2$

b) $(P \vee \neg r) \wedge (q \vee \neg s)$ $2^4 = 16$

c) $q \vee P \vee \neg s \vee \neg r \vee \neg t \vee u$ $2^6 = 64$

d) $(P \wedge r \wedge t) \leftrightarrow (q \wedge t)$ $2^4 = 16$